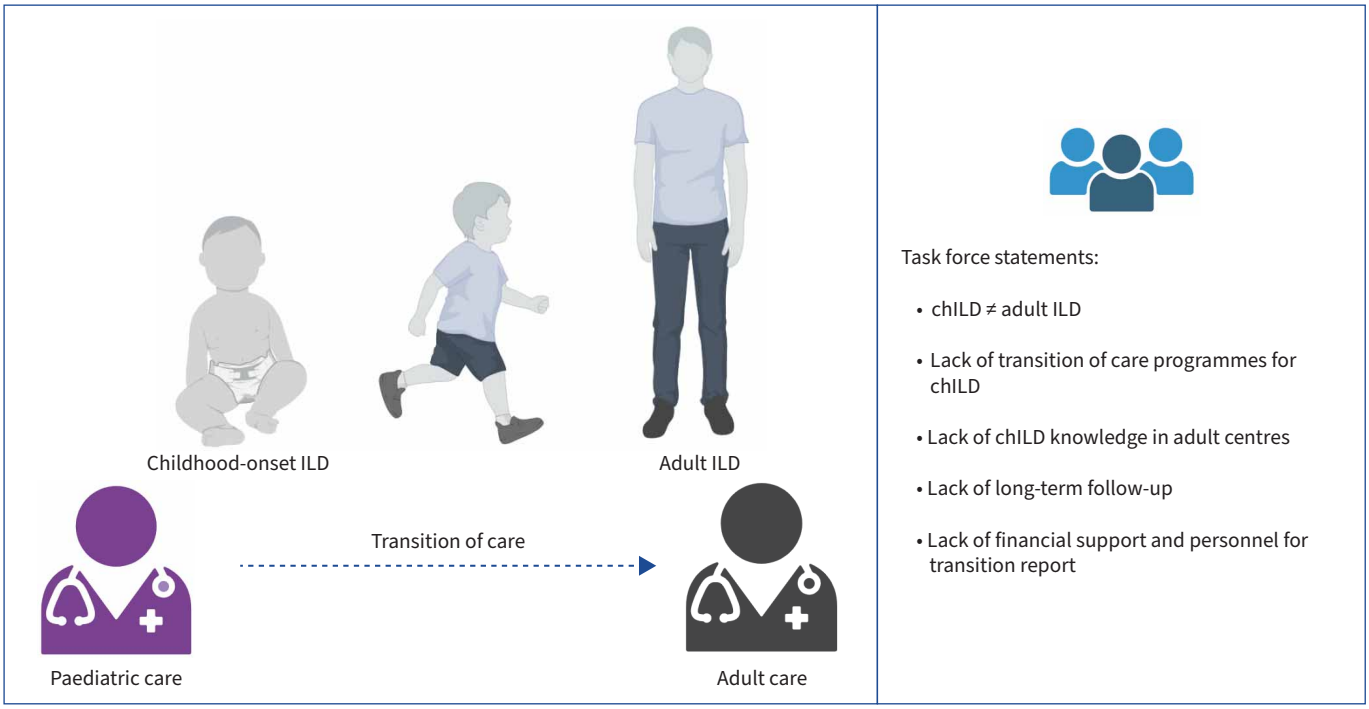
ERS statement on transition of care in childhood interstitial lung diseases

Petr Pohunek, Effrosyni Manali, Susanne Vijverberg, Julia Carlens, Felix Chua, Ralph Epaud, Carlee Gilbert, Matthias Griese, Bulent Karadag, Eitan Kerem, Václav Koucký, Nadia Nathan, Spyridon Papiris, Suzanne Terheggen-Lagro, Lukáš Plch, Alba Torrent Vernetta and Andrew Bush



GRAPHICAL ABSTRACT Overview of this European Respiratory Society statement on transition of care in childhood interstitial lung diseases (chILD).

https://doi.org/10.1183/13993003.02160-2023

Downloaded from https://publications.ersnet.org on March 28, 2026 by guest. Please see licensing information on first page for reuse rights.

ERS statement on transition of care in childhood interstitial lung diseases

Petr Pohunek [1], Effrosyni Manali[2], Susanne Vijverberg [3,4], Julia Carlens [5], Felix Chua[6,7], Ralph Epaud [8,9,10,11], Carlee Gilbert [12,13], Matthias Griese [14], Bulent Karadag [15], Eitan Kerem [16], Václav Koucký [1], Nadia Nathan [17], Spyridon Papiris[2], Suzanne Terheggen-Lagro [4], Lukáš Plch [18,19], Alba Torrent Vernetta [20,21] and Andrew Bush [22]

[1]Paediatric Pulmonology, Paediatric Department, 2nd Faculty of Medicine and University Hospital Motol, Prague, Czech Republic. [2]2nd Pulmonary Medicine Department, General University Hospital, Attikon, Medical School, National and Kapodistrian University of Athens, Athens, Greece. [3]Pulmonary Medicine, Amsterdam UMC, University of Amsterdam, Amsterdam, The Netherlands. [4]Pediatric Pulmonology and Allergy, Emma Children's Hospital Amsterdam UMC, University of Amsterdam, Amsterdam, The Netherlands. [5]Department of Pediatrics, Pediatric Pulmonology, Allergology and Neonatology, Hannover Medical School, Hannover, Germany. [6]Interstitial Lung Disease Unit, Royal Brompton and Harefield Hospitals, London, UK. [7]The Margaret Turner Warwick Centre for Fibrosing Lung Diseases, Imperial College London National Heart and Lung Institute, London, UK. [8]Centre Hospitalier Intercommunal de Créteil, Service de Pédiatrie Générale, Créteil, France. [9]Centre des Maladies Respiratoires Rares (RESPIRARE), CRCM, Créteil, France. [10]Fédérations Hospitalo-Universitaires (FHU) Role of SENEscence in Chronic Diseases (SENEC), Créteil, France. [11]University Paris Est Créteil, INSERM, IMRB, Créteil, France. [12]Institute of Population Health, The University of Liverpool, Liverpool, UK. [13]ChILD Lung Foundation, UK. [14]Department of Pediatric Pneumology, Dr von Hauner Children's Hospital, Ludwig-Maximilians-University, German Center for Lung Research, Munich, Germany. [15]Marmara University School of Medicine, Division of Pediatric Pulmonology, Istanbul, Turkey. [16]Department of Pediatrics and CF Center, Hadassah Hebrew University Medical Center, Jerusalem, Israel. [17]Pediatric Pulmonology Department and Reference Centre for Rare Lung Diseases RespiRare, INSERM UMR_S933 Laboratory of Childhood Genetic Diseases, Armand Trousseau Hospital, Sorbonne University and APHP, Paris, France. [18]University Campus Library, Masaryk University, Brno, Czech Republic. [19]Department of Educational Sciences, Faculty of Arts, Masaryk University, Brno, Czech Republic. [20]Pediatric Allergy and Pulmonology Section, Department of Pediatrics, Vall d'Hebron Hospital Universitari, Vall d'Hebron Barcelona Hospital Campus, Universitat Autònoma de Barcelona, Barcelona, Spain. [21]CIBERER, Centro de Investigación en Red de Enfermedades Raras, Instituto de Salud Carlos III, Madrid, Spain. [22]National Heart and Lung Institute, Imperial College, Royal Brompton and Harefield NHS Foundation Trust, London, UK.

Corresponding author: Petr Pohunek (petr.pohunek@lfmotol.cuni.cz)

Shareable abstract (@ERSpublications)
Childhood interstitial lung diseases are rare. Continuous care and follow-up from infancy and childhood to adult life is crucial to ensure optimal treatment. This @ersociety statement provides the latest insights on transition of care in #chILD https://bit.ly/44UOLPq

Cite this article as: Pohunek P, Manali E, Vijverberg S, *et al*. ERS statement on transition of care in childhood interstitial lung diseases. *Eur Respir J* 2024; 64: 2302160 [DOI: 10.1183/13993003.02160-2023].

Copyright ©The authors 2024. For reproduction rights and permissions contact permissions@ersnet.org

Received: 1 Dec 2023
Accepted: 19 April 2024

Abstract

Interstitial lung diseases (ILD) are a heterogeneous group of rare diffuse diseases affecting the lung parenchyma in children and adults. Childhood interstitial lung diseases (chILD) are often diagnosed at very young age, affect the developing lung, and can have different presentations and prognosis compared to adult forms of these diseases. In addition, chILD in many cases may apparently remit, and have a better response to therapy and better prognosis than adult ILD. Many affected children will reach adulthood with minimal activity or clinical remission of the disease. They need continuing care and follow-up from childhood to adulthood if the disease persists and progresses over time, but also if they are asymptomatic and in full remission. Therefore, for every chILD patient an active transition process from paediatric to adult care should be guaranteed. This European Respiratory Society (ERS) statement provides a review of the literature and current practice concerning transition of care in chILD. It draws on work in existing transition care programmes in other chronic respiratory diseases, disease-overarching transition-of-care programmes, evidence on the impact of these programmes on clinical outcomes, current evidence regarding long-term remission of chILD as well as the lack of harmonisation between the current adult ILD and chILD classifications impacting on transition of care. While the transition system is well established in several chronic diseases, such as cystic fibrosis or diabetes mellitus, we could not find sufficient published evidence on transition systems in chILD. This statement summarises current knowledge, but cannot yet provide evidence-based recommendations for clinical practice.

Downloaded from https://publications.ersnet.org on March 28, 2026 by guest. Please see licensing information on first page for reuse rights.

Relevance for patients

Transition of patients with chILD from paediatric to adult care is of crucial importance if the long-term course of the disease and its prognosis is to be optimal. This ERS task force document thoroughly maps the current state of transition and the gaps in the present service and provides a basis for improving the present fragmented and inconsistent protocols and the overall care for these patients throughout the life-course.

Introduction

Chronic interstitial lung diseases (ILD) are a group of rare diffuse lung diseases that affect the lung parenchyma, leading to varying types of inflammation, alveolar filling and subsequent fibrosis. This leads often to loss of lung capacity, decreased lung compliance, deterioration of diffusion capacity and to the gradual development of dyspnoea, respiratory failure, disability and even death [1]. Many of these diseases result from genetic defects, and originate in the neonatal period, early infancy or childhood [2, 3]. Course and prognosis are rather variable, ranging from lethal neonatal respiratory distress syndrome to "fibrotic" ILD continuing from childhood into adulthood. In some diseases, there may be a potential for remission, either spontaneous or induced by treatment; however, some of these diseases can persist or be progressive over time. If remission occurs, the patients often have a residual impairment of lung function and there may be a significant lifelong risk of relapse even after years of quiescence. Children with interstitial lung diseases (chILD) always need centralised paediatric specialist care because of the rarity (prevalence up to 46/1 000 000 children) and diversity (>200 entities) of these conditions [4]. It is very important to provide these patients with long-term follow-up if they attain apparent remission during childhood. There may be long-term side-effects of the treatment, especially oral corticosteroid therapy [5]. Furthermore, there may be important genetic implications for future children, and it is important that as we become aware of even more genetic entities, the possible underlying genetic basis for an undiagnosed chILD is reviewed regularly. Thus, when reaching adulthood, these patients must be transitioned to the care of adult ILD departments to guarantee uninterrupted specialised monitoring and follow-up. As with other chronic diseases in childhood, for chILD it is critical to bring disciplines together and studying paediatric diseases over the whole life-course. Disparities between and within countries need to be addressed, as well as a lack of clinical trials [6].

Transition protocols have already been developed for a few chronic diseases (*e.g.* cystic fibrosis, diabetes mellitus, cardiac and metabolic diseases, sickle cell anaemia and inflammatory bowel disease), and these are still being improved [7–13]. Much less work has been done about transition in rare respiratory diseases including chILD. However, from other chronic diseases we know that insufficient support during transition to adult care or discontinuity of care are often associated with severe deterioration of health status [14]. The highest risk of being lost to follow-up is seen in children with less severe disease or with some degree of remission/stability. They may seek medical care again only when the disease has symptomatically relapsed or deteriorated. This may be too late to prevent avoidable permanent damage of the lungs and thus a worse prognosis.

Scope

The aim of this European Respiratory Society (ERS) task force (TF-2021-14) was to produce a statement on transition of care in chILD based on a systematic review of the literature, information obtained from previous work done during the European Cooperation in Science and Technology (COST) Action 16125 (European Network for Translational Research in Children's and Adult Interstitial Lung Disease (ENTeR-chILD)), as well as observations from clinical practice and experience of the members of the task force. This statement describes the current evidence on best practice of transition of care in children with chronic ILD; contrasts this with current clinical practice; and identifies areas for future research concerning transition of care in chILD.

Methods

The current statement was developed by the ERS task force on transition of care in chILD. This multidisciplinary task force consisted of 17 members (including three Early Career Members), 11 experts in chILD, three experts in adult ILD, one epidemiologist, one psychologist/patient representative and a librarian. The work was supported by an ERS methodologist. Members were selected based on their expertise and to ensure wide geographical representation and gender balance. All members of the task force disclosed their conflicts of interest before initiation of the project and upon submission of the manuscript. At the first meeting, the task force agreed on the overall approach. A list of five clinically important questions was compiled by the task force members (table 1); one PICO (Population, Intervention, Comparator, Outcomes) question and four narrative or non-PICO questions. Members were assigned to working groups within the task force addressing the different questions. The work of each working group was predominantly by email and online meetings. Two hybrid (physical and online)

Downloaded from https://publications.ersnet.org on March 28, 2026 by guest. Please see licensing information on first page for reuse rights.

TABLE 1 Prioritised clinical questions underlying this statement

Q1 (PICO)	Does participation of children with a chronic respiratory disease in a care transition programme lead to better outcomes?
Q2	Which care transition programmes are currently used in clinical practice for children with ILD?
Q3	In which other diseases are transition of care programmes used in clinical practice?
Q4	Is there evidence of long-term remission of chILD?
Q5	Does the current definition of chILD *versus* adult ILD influence transition of care?

PICO: Population, Intervention, Comparator, Outcomes; ILD: interstitial lung disease; chILD: childhood ILD.

meetings of the task force members were held during the annual ERS Congresses: in Barcelona on 3 September 2022 and in Milan on 11 September 2023. To add to published evidence, the working groups also identified ongoing studies, other sources (*e.g.* survey results from the COST Action) and clinical experiences that were relevant to the respective topics. This ERS statement combines an evidence-based approach, clinical expertise of task force members, systematic search of the literature and critical discussions from virtual meetings. The statement summarises relevant literature and current practice, but does not make any recommendations.

Systematic literature searches

To answer the PICO question (Does participation of children with a chronic respiratory disease in a care transition programme lead to better outcomes?), a systematic literature search was executed by the specialist librarian (L. Plch). A protocol was registered at PROSPERO (www.crd.york.ac.uk/prospero; identifier number CRD42022320036). Systematic searches were conducted on 12 February 2022 in MEDLINE/PubMed, EMBASE and Cochrane. The PICO and full search strategies are presented in the supplementary material. The main inclusion criteria were based on the PICO (supplementary material). The main exclusion criteria were articles published in languages other than English and article types different from research articles and studies published >20 years ago. Search results were uploaded onto the Rayyan platform (https://rayyan.qcri.org). Two authors (P. Pohunek, E. Manali) independently screened the titles and abstracts identified in the search using the inclusion criteria. Once potentially relevant studies had been identified, the same two reviewers independently screened the full-text publications to determine eligibility for inclusion. Disagreements were settled by consensus of the two reviewers and a third reviewer (S.J. Vijverberg). We recorded the reasons for excluding search results that were not eligible once screened by full text. Data from eligible studies were extracted independently by two reviewers (P. Pohunek, E. Manali). Data extracted include study design, number of patients, patient characteristics (age, gender, condition), details about the intervention (duration of follow-up, disciplines involved, measures assessed) and outcomes. Results of our systematic review was reported according to the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) checklist. Our PRISMA diagram is shown in figure 1.

For the remaining non-PICO questions presented in the narrative form (questions 2–5), more narrow *ad hoc* searches were used based on each question's topics. The *ad hoc* searches were performed by the authors of the individual subchapters based on the keywords relevant to the topic of the question. They searched the PubMed and personal archives and checked the reference list for any missed important manuscripts.

Development of the statement

The first outline of the document was prepared by the core working group of the task force (P. Pohunek, E. Manali, S.J. Vijverberg, A. Bush). The data from the systematic review and the outline were reviewed during the hybrid task force meeting. Individual questions were then assigned to the subgroups of the task force members who prepared the relevant chapters. Based on this, the next draft of the document was prepared and circulated to all the task force members for comments and modified accordingly.

Results

Question 1 (PICO): Does participation of children with a chronic respiratory disease in a care transition programme lead to better outcomes?

To assess the added value of participation in care transition programme we performed a systematic literature search. The systematic search identified 946 publications (MEDLINE/PubMed n=225, EMBASE n=332 and Cochrane n=389); 27 full-text articles were retrieved for further evaluation of eligibility. 10 studies were included in the qualitative synthesis. Results are depicted in table 2 and refer almost exclusively to patients with cystic fibrosis, although some studies also included nonrespiratory disorders

Downloaded from https://publications.ersnet.org on March 28, 2026 by guest. Please see licensing information on first page for reuse rights.

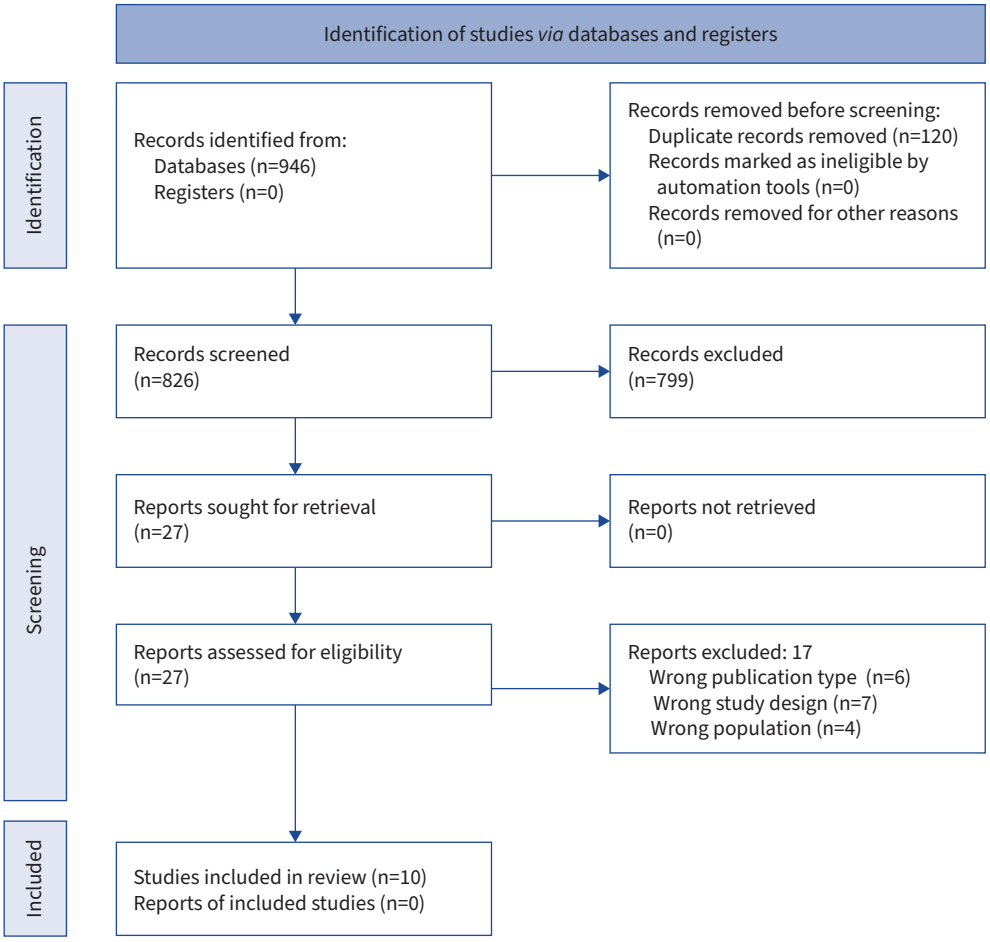


FIGURE 1 Preferred Reporting Items for Systematic Review and Meta-Analysis diagram of study selection.

such as inflammatory bowel disease and type-1 diabetes mellitus. This highlights the paucity of data about the transition of chILD patients [1, 15–23]. Based on the aforementioned studies, the participation of children with a chronic disease in a transition care programme enables a more successful transfer by lowering pre-transfer anxiety and significantly improving transition competence and readiness, self-efficacy and independence, as well as satisfaction with school/educational services. Overall quality of life and health-related outcomes such as lung function and body mass index do not worsen, but remain stable in the majority of studies, and seem to improve in a few. The benefits of the transition care programmes are observed not only in patients, but also in their parents.

Question 2: Which care transition programmes are currently used in clinical practice for children with ILD?

The literature search narrowed to chILD and transition programmes from paediatric to adult care revealed seven papers. Two papers addressed nonrelevant topics; two addressed only a narrow spectrum of diseases, mainly with regard to the long-term prognosis. Two others concentrated more on the diagnostic aspects and the need for development of guidelines. Therefore, we found only limited data on the transition process of patients with chILD from paediatric to adult care systems that are already used in clinical practice. There is one paper [25] which deals with the transition process of chILD in general. It reports outcomes from the COST Action 16125 (working group 5, dedicated to the transition process). In a survey among 39 centres from 19 European and two non-European countries, which was performed by this working group, there were no standardised transition protocols and no established transition system for children with ILD. They identified only three (11%) centres having a standard operation protocol for transition and nine (32%) centres using a standardised medical report form that accompanied the patient during the whole transition process. Nine (32%) centres organised a meeting between paediatric and adult teams without patient/parents; only two (18%) adult centres had an established system of long-term

Downloaded from https://publications.ersnet.org on March 28, 2026 by guest. Please see licensing information on first page for reuse rights.

TABLE 2 Effectiveness of care transition programmes for chronic respiratory diseases according to the studies for PICO (Population, Intervention, Comparator, Outcomes) question 1 based on the systematic research and the strategy described in figure 1 and supplementary tables S1 and S2, respectively

First author, year [reference]	Disease(s) studied	Description	Number of patients and age (years) at transfer	Disciplines involved	Intervention	Outcome assessed	Main conclusion
Huang, 2014 [15]	Inflammatory bowel disease, CF, type 1 diabetes	Controlled intervention study	n=81 Intervention: n=40, 17 male, median age 17 (range 12–20) years Control: n=41, 20 male, median age 17 (range 12–19) years	Nurse, scheduler, physician	Technology-based, transition intervention to improve disease management, health-related self-efficacy and patient-initiated communication programme	Disease management skills, health related self-efficacy, patient-initiated healthcare communications	Outcomes in adolescents with chronic disease improved significantly following a generic, technology-based intervention, in a cost-effective manner
Schmidt, 2016 [16]	Inflammatory bowel disease, CF, type 1 diabetes	Controlled intervention study	n=325 53.8% male Mean±SD age 16.8±1.74 years	The treating physicians informed the patients about the programme	Transition workshop of 2 days with 8 modules of 60–90 min	Health-related transition competence, self-efficacy, patient activation and satisfaction with healthcare, QoL	Highly significant improvement in transition competence as an overall scale and on all subscales, a significant improvement of self-efficacy, satisfaction with school/educational services, between study intake and follow-up, no difference in QoL, patient activation, general satisfaction with care
Bourgeois, 2021 [17]	CF	Pre–post study with no control	n=97 47.4% male Mean±SD age 18.8±0.6 years	Multidisciplinary team members from paediatric and adult care teams (nurses, physicians, physiotherapists, psychologists)	A 10-year-old formalised process of transitioning patients from a paediatric to an adult CF centre	Evolution of pulmonary function, nutritional status, pulmonary colonisation, antibiotic course, days of hospitalisation and evolution of outpatient visits	Transitioning from the paediatric centre to an adult centre did not worsen clinical outcomes, including lung function and nutritional status
Welsner, 2019 [18]	CF	Pre–post study with no control	n=39 64% male Mean±SD age 29±9.3 years	Treating paediatrician, CF nurses and psychosocial workers, adult CF specialist	Short-term transfer programme	Lung function, bacterial colonisation, nutritional status, out and inpatient contacts, clinical events (*e.g.* lung transplantation, pneumothorax, haemoptysis)	A short-term transfer programme facilitates a successful transfer

Continued

https://doi.org/10.1183/13993003.02160-2023

Downloaded from https://publications.ersnet.org on March 28, 2026 by guest. Please see licensing information on first page for reuse rights.

TABLE 2 Continued

First author, year [reference]	Disease(s) studied	Description	Number of patients and age (years) at transfer	Disciplines involved	Intervention	Outcome assessed	Main conclusion
Peeters, 2019 [19]	CF	Controlled intervention study	27 transition clinic and 19 direct handover Mean±SD age at transfer 18.31±0.618 years *versus* 18.36±0.633 years Male 48.1% *versus* 52.6%	Professionals from both paediatric care and adult care	Transition clinic programme *versus* direct handover care	Transfer experience, self-management, health care use, clinical outcomes	Transition clinic had several perceived benefits and showed positive trends in transfer experiences and satisfaction, but no difference in health-related outcomes
Skov, 2018 [20]	CF	Observational cohort (interventional no control?)	n=40 15 female, mean age 15.7 (13–18) years and 25 males mean age 14.6 (12–17) years	Paediatric CF experts, nurses, physiotherapist, physiologist, psychologist, nutritionist, social worker	A well-structured transition programme for adolescents with CF starting at the age of 12 years	Feasibility, acceptability, practicality and implementation and efficacy assessment (patient-reported outcomes, clinical parameters)	An early start of the transition programme was feasible and successfully introduced and implemented; self-evaluated transition readiness increased significantly, overall quality of life, lung function and BMI remained stable
Chaudhry, 2013 [21]	CF	Controlled intervention study	n=91 50 male Age 30.8±9.3 years Age at transfer of care: participants 19.1±2.1 years (24 male), nonparticipants 21.6±6.0 years (26 male)	Paediatric and adult pulmonologist, transition coordinator, social worker, psychologist	Application of the transition programme of a specific institution	Anxiety before/after, satisfaction before/after, perceived health status before/after, independence before/after	Patients participating in the programme had higher satisfaction, perceived health status and independence
Tuchman, 2013 [22]	CF	Retrospective matched cohort study	Transition-negative: 4028, 53.7% female, age 24.26 years Transition-positive: 703, 51.2% female, age 21.24 years	Not clearly stated (those who participated at the paediatric and adult CF programme?)	Transition from paediatric to adult programme	Rate of change in outcome measures per year after transition (FEV_1 % pred, FVC % pred, BMI, number of hospitalisations, number of home *i.v.* antibiotic events	There was a less rapid decline in pulmonary function 2 years after the transfer from paediatric to adult care when compared with those who remained at paediatric care; there were no observed changes in rates of change in FVC, BMI, annual hospitalisations or annual home *i.v.* antibiotic events 2 years after

Continued

https://doi.org/10.1183/13993003.02160-2023

Downloaded from https://publications.ersnet.org on March 28, 2026 by guest. Please see licensing information on first page for reuse rights.

TABLE 2 Continued

First author, year [reference]	Disease(s) studied	Description	Number of patients and age (years) at transfer	Disciplines involved	Intervention	Outcome assessed	Main conclusion
Middour-Oxler, 2022 [23]	CF	Retrospective cohort study	Semi-formal: n=29, 24.1% male, mean age 20.9 years Formal: n=27, 49.1% male, mean age 18.6 years	Multidisciplinary paediatric and adult CF team	Formalised *versus* non-formalised transition programme	Self-perceived transition-related anxiety, transition readiness and satisfaction in subjects who participated in a formalised transition programme	This study supports that participation in a formalised transition programme was associated with significantly lower pre-transfer anxiety and higher post-transition satisfaction in individuals with CF
Craig, 2007 [24]	CF	Cross-sectional survey	Pre-transitional: n=45, 64% female, age 16.2±1.5 years, n=45 parents Post-transitional: n=27, 44% female, 20.9±1.8 years, n=20 parents	Paediatric and adult services staff, transition coordinator	Those who participated at a greater number of steps at the transition programme *versus* the others	Participants' concerns about adult healthcare, patient participation in transition, patient satisfaction with transition, CF-related QoL	Most people were satisfied with the transition programme and post-transition patients and parents who completed more steps in the transition describe the overall transition programme more positively; there was no evidence for a relationship between amount of participation in transition programme and measures of disease severity and QoL

CF: cystic fibrosis; QoL: quality of life; BMI: body mass index; FEV_1: forced expiratory volume in 1 s; % pred: % predicted; FVC: forced vital capacity; *i.v.*: intravenous.

https://doi.org/10.1183/13993003.02160-2023

Downloaded from https://publications.ersnet.org on March 28, 2026 by guest. Please see licensing information on first page for reuse rights.

follow-up of patients with remitted chILD. The age of transition from paediatric to adult healthcare was mostly 18 years, ranging from 16 to 21 years. There was a high variability in the answer to the question about how long before the actual transition the transition process starts. This varied between 3 months to 4 years, with 1 year being most frequent. The availability of ILD multidisciplinary teams involved in care and transition were rather variable between countries. Only 20 (51%) of all responding centres reported availability of an established multidisciplinary team. Most frequently, pulmonologists/paediatric pulmonologists, radiologists and pathologists were involved. Only four (14%) paediatric centres routinely organised a pre-transition meeting with the patient, parents and both the paediatric and adult teams; nine (32%) centres had a meeting between paediatric and adult teams without patient/parents; five (18%) paediatric centres organised a meeting of the paediatric team with patient/parents. No centre reported an established system of long-term follow-up of children with chILD specific for early age (for example, neuroendocrine cell hyperplasia of infancy (NEHI), pulmonary interstitial glycogenosis).

Koucký *et al.* [25] suggested a model procedure for the transition of care. The patient would be re-assessed in detail during the last year of follow-up at the chILD centre. Based on the practices of the task force members, a thorough evaluation of clinical and functional status is useful, and radiology (chest high-resolution computed tomography (HRCT)) will usually be considered depending on the specific diagnosis and activity of the disease. Finally, a standardised transition report was suggested. There were no other articles found detailing individual chILD transition programmes and evaluating their efficacy with respect to patients' disease management skills, patient access to and satisfaction with healthcare, self-efficacy, risk of loss to follow-up and risk of missing disease progression.

In the current clinical practice of the task force members, the timing and extent of diagnostic re-evaluation during the transition process is determined on a case-by-case basis, in the absence of any definitive evidence. Although some forms of ILD are common to both paediatric and adult pulmonary practice, dissimilarities between paediatric and adult disease classifications as well as evolution of the condition between the time of the original paediatric diagnosis and subsequent transition to adulthood are among the reasons why clinicians re-evaluate the patients on an *ad hoc* basis.

Task force members indicated that usually the process incorporates input from the paediatric and adult clinical teams including specialist nurses and respiratory support staff, as well as the patient and their family or carers. The task force members believe that clinical considerations, such as the history of the condition, its management up to that point and response to therapy may be important factors influencing the decision whether to re-evaluate at transition.

Discussion within the task force regarding the current practice identified the most commonly used components of a general clinical re-evaluation plan: 1) blood investigations including extended autoantibody panel, serology for hypersensitivity pneumonitis (IgG against a number of organic stimuli such as avian and fungal antigens) and repeat genetic testing for entities likely due to a known and not yet tested condition; 2) physiological testing such as detailed lung function testing (including static lung volumes, transfer factor of the lung for carbon monoxide), overnight oximetry, 6-min walk test and possibly oesophageal impedance–pH assessment; 3) thoracic radiological imaging with HRCT as the key modality; and 4) echocardiography (baseline cardiac chamber dimensions, biventricular function and estimate of pulmonary artery pressure). With respect to the radiation dose, the HRCT is done on a case-by-case basis. The cumulative radiation burden has to be taken into account; however, with modern imaging technologies (ultrasound, magnetic resonance imaging, ultra low-dose computed tomography) this may become less important [26]. A list of minimal content of the transition report is presented in table 3 [25]. No published evidence-based validated list could be found in the literature.

Question 3: In which other diseases are transition of care programmes used in clinical practice?

Most children and young people with chronic illnesses will survive into adulthood, and would optimally transition to an adult model of care. A literature search revealed 1748 papers with numerous transition protocols in different pathologies, such as rheumatological diseases, congenital heart disease, inflammatory bowel disease, chronic neurological disorders, diabetes mellitus and patients on long-term ventilation [8, 10, 12, 13, 27, 28]. Most of these patients have complex conditions requiring multidisciplinary medical care teams and are classified as "youth with special healthcare needs" [14].

Based on the literature and task force members' personal practice, an important goal of transition is to empower patients and families, aiming at increasing autonomy, patient participation, awareness and consciousness, as well as the development of relevant psychosocial skills. It is also crucial to ensure an organised process to facilitate transition preparation, transfer of care and integration into adult-centred

Downloaded from https://publications.ersnet.org on March 28, 2026 by guest. Please see licensing information on first page for reuse rights.

TABLE 3 A sample proposal for possible minimal content of the transition report

Age at first clinical symptoms	
Age at diagnosis of chILD	
Precise diagnosis and estimation of prognosis	
Detailed data	Perinatal history
	Other morbidities: severe infections, need for ventilation support, *etc*.
	Involvement of other organs
	Vaccinations
	Exposure to external risk factors (smoking, allergens, chemicals and other toxins, environmental pollution, occupation exposure in adolescents, school and work), past and present
	School, (considered) job preferences with regard to possible exposure to external risk factors
	Familial occurrence of risk factors (a full genogram should be prepared); miscarriage and stillbirth, death in neonatal/infant period; unexplained respiratory distress, sepsis, as well as occurrence of ILD in first- and second-degree relatives, especially in combination with haematopoietic disorders, liver, brain or thyroid disease, malignancies before the age of 50 years
	Social and psychological history (social background, cooperation, adherence to the treatment regime, possible psychological problems, understanding of the disease and its acceptance). An official social worker's report may be indicated in some patients
Results of previous examinations relevant to the diagnosis and their development over time	Detailed physical examination, including anthropometric measurements, lung auscultation, digital clubbing, any chest deformity, hepato/splenomegaly, skin changes, ophthalmological examination
	Result of cardiological examination, if relevant; indirect signs of pulmonary hypertension, systemic hypertension
	Imaging, especially HRCT
	Lung function (spirometry, body plethysmography, diffusion capacity, objective assessment of exercise tolerance and its evolution over time, cardiopulmonary exercise testing, if available)
	Past CBC and other blood test abnormalities, and most recent CBC and biochemistry, and serological tests and autoimmune tests
	Date and details of genetic testing
	Bronchoscopic examination including results of bronchoalveolar lavage: differential count of leukocytes and their immunophenotyping, presence of haemosiderin-laden macrophages, lipid-laden macrophages, CD1a-positive cells
	Result of lung biopsy: age at examination, method, and anatomical site of biopsy (side and lobe), microscopic finding including usual staining and histopathological classification
	Others: densitometry, bone metabolism, oral glucose tolerance test, ophthalmological examinations
Clinical course of the disease	Occurrence of acute pulmonary exacerbations, their triggers, method of treatment, development of lung function and laboratory findings with regard to the progression of lung disease
Current and past treatments	Pharmacological treatment (duration and response): systemic corticosteroids, other immunosuppressive agents, azathioprine, mycophenolate mofetil, cyclophosphamide, hydroxychloroquine, azithromycin, possibly antifibrotics in a clinical study (active substance, dose, duration of treatment)
	Side-effects of medication
	Nonpharmacological treatment: home oxygen therapy, respiratory physiotherapy, other treatment methods (diet…)
Were patients included in a registry?	If so, transfer to adult register
Patient's perspective	Important issues of the disease as perceived by the patients themselves
Summary	

Information from [25]. chILD: childhood interstitial lung disease; ILD: interstitial lung disease; HRCT: high-resolution computed tomography; CBC: complete blood count.

healthcare. Without such a process, there is a big risk that children and young people, even with complex ongoing treatment needs, will never transition, and be lost to follow-up [29–31]. Some individual centres and different organisations have developed transition programmes to achieve these goals. In 2011, updated in 2018, the American Academy of Pediatrics, with the American Academy of Family Physicians and the American College of Physicians, published a clinical report on healthcare transition that included a process

Downloaded from https://publications.ersnet.org on March 28, 2026 by guest. Please see licensing information on first page for reuse rights.

for transition preparation, planning, tracking and follow-through [32, 33]. Afterwards, a structured clinical approach called the Six Core Elements of Health Care Transition was developed for all children and young people [34]. An important part of the process is talking to children and young people using the acronym HEADSS (home, education, activities/employment, drugs, suicidality, sex) [35].

The literature highlights the need for structured elements in a healthcare transition programme: a written transition policy, and a transition plan, which is a working document that records findings of assessments, describes the progress in transition and outlines planned actions. A very good example of this is the Ready Steady Go project in the United Kingdom (UK), which has been developed as a generic transition plan, and comprises assessment about knowledge and skills over time [36]. This programme is being used to develop increasing numbers of disease specific versions. Another example is the Got Transition (United States of America) Six Core Elements of Health Care Transition 2.0 programme (www.GotTransition.org). These programmes help to facilitate the transition process for the young person and try to involve the patient in the process and help in various settings and conditions. In addition, the UK National Institute for Health and Care Excellence has published excellent recommendations for transition to adult services [31].

Most transition programmes start in early adolescence and continue into early adulthood; however, the fact that transition will occur is usually mentioned much earlier, often at diagnosis. Usually, these programmes are provided in three phases: pre-transition, where the transition process and transfer policy is introduced to the patient and family; the transition phase, where the needs of the adolescent are assessed, he/she is enabled to manage his/her own health supported by education and counselling, and also includes the actual transfer to adult care; and the post-transition phase, when the continuity of care is monitored.

The task force members believe, based on their practice and current literature, that for most patients, the appropriate time for implementing specific stages of transition planning is determined by age, whereas for others, it may be modified based on developmental considerations [7, 8]. As this is a vulnerable time for young adults, and is also a challenging period for medical teams, patient care during transition would optimally be individualised.

Transition planning is a multidisciplinary process whereby different disciplines are involved in order to address different areas of need. Physicians and nurses usually discuss and assess medical aspects, disease knowledge, management tasks and health-related self-efficacy; psychologists are suited to target issues of nonadherence to treatment, and behavioural or emotional factors; social workers can help patients by connecting them with resources that may facilitate transfer. Recently, the crucial role of adult clinicians in accepting and partnering with young adults has been emphasised as a professional education and training challenge. For the most part, no published data has been found regarding diagnostic re-evaluation in the transition literature, but there are some (non-evidence-based) exceptions [37].

Transfer usually occurs at a time of disease stability. The task force members agreed that in their practice, supported by current literature, it was found essential to provide a report assessing readiness for transfer, as well as the important information that summarises all relevant data about the patient's medical history (an example being shown in table 3).

Additional data that are likely to be included in the transfer documentation include family history, bespoke treatment plans and contacts, and the patient's health education history and assessment of his/her understanding regarding health conditions, treatments, and prognosis with particular attention to entry into adult life [8, 12, 13, 28, 38].

The evidence on healthcare transition outcomes remains limited, but there are some evaluation studies that document beneficial outcomes of a structured transition approach in terms of quality of care and in terms of service use and patient and family experience. A systematic review identified 43 studies, of which two-thirds revealed statistically significant positive outcomes. The most commonly reported outcome was improvement in adherence to care, followed by improved perceived health status, quality of life and self-care skills [39].

In 2016, a Cochrane review [40] evaluated the effectiveness of interventions designed to improve the transition of care for adolescents from paediatric to adult health services. The few available pieces of evidence, which covered a limited range of interventions in a limited number of clinical conditions, showed improvement in patients' knowledge of their condition, and improvement in self-efficacy and confidence. Nevertheless, it concluded that further research is needed for rigorous evaluation of models of transitional care, reporting on clinical outcomes and patient-reported outcomes with longer-term follow-up.

Downloaded from https://publications.ersnet.org on March 28, 2026 by guest. Please see licensing information on first page for reuse rights.

A recent study showed in a small sample that the common experiences included lack of transition preparation and planning, the need for learning how to adapt to adult services and a changing healthcare scene. Additionally, the subject of how to include the parent as a continued advocate for the young person was discussed. Among the weaknesses identified in the current transition process were the lack of a specific transition coordinator, the generally haphazard timing of first discussion about transition and the lack of involvement of psychologists and social workers [41].

Specific characteristics of the different transition programmes for well-studied chronic diseases of childhood other than chronic respiratory diseases are included in table 4 [8, 10, 12, 13, 27, 28]. Regarding chronic respiratory diseases, the nature and structure of the process are generally well described in cystic fibrosis care, where a relatively large number of patients transition each year. Cystic fibrosis is one of the diseases with best-studied transition pathway in the literature (table 1) [7, 43]. The task force members considered that some aspects could potentially be successfully transferred to chILD transition programmes. However, there are gaps in knowledge and recommendations regarding the structure of diagnostic re-evaluation for chILD which encompasses a more heterogeneous group of diseases than cystic fibrosis. There are also organisational issues: for a rare disease like chILD it will not be feasible for every centre to have a dedicated chILD disease coordinator, so a shared approach with other diseases is probably necessary. Lastly, there may be difficulties with regard to selection of the appropriate adult expert centre, especially for those entities of chILD not well-known to adult ILD experts, an issue better addressed in question 5 (Does the current definition of chILD *versus* adult ILD influence transition of care?).

Question 4: Is there evidence of long-term remission of chILD?

There is still a big knowledge gap on natural disease course, disease progression and treatment induced or spontaneous remission of chILD. We identified a limited number of studies on the long-term evolution of various chILDs in our literature search. Unfortunately, these papers provide no information on the transition process itself and do not evaluate its prognostic impact. There are only few studies reporting health status of patients transitioning from childhood to adulthood [41]. Moreover, these data are significantly biased, as those with unsuccessful transition are unlikely to be reported (loss to follow-up).

Although there is no specific definition of remission of chILD, based on studies and expert opinion it could be postulated that remission consists of four dimensions [44]:

- clinical signs and symptoms;
- pulmonary function (spirometry, body plethysmography, diffusion capacity of the lung for carbon monoxide (D_{LCO}), exercise tolerance tests);
- imaging (HRCT scan of the chest);
- pathology, cytology (results of tissue biopsy and bronchoalveolar lavage abnormalities).

TABLE 4 Characteristics of care transition programmes for well-studied chronic diseases of childhood other than respiratory disease

First author, year [reference]	Disease	Age of first joint consultation	Disciplines involved
Got Transition® programme [42]	Generic	Starts 12–14 years Ends 18–23 years	Multidisciplinary team
Nagra, 2015 [36]	Generic	Starts 11–12 years Ends 16–18 years	Not specified
Calvo, 2015 [28]	Rheumatological diseases	Starts 10–14 years Ends 18–24 years	Paediatric and adult rheumatologist, psychologist, others not specified
Buschur, 2017 [8]	Diabetes mellitus	Starts 14 years Ends 18 years	Paediatric and adult endocrinologist, psychologist, social worker
Philpott, 2019 [12]	Inflammatory bowel disease	Starts 15 years Ends 21 years	Physicians, nurses, social workers, psychologists, medical assistants
Bolger, 2017 [38]	Cerebral palsy	19–22 years	Paediatric neurologist, neurosurgeon, psychologist, orthopaedic surgeon, orthotist, occupational therapist, social worker, speech therapist
Moons, 2021 [13]	Congenital heart disease	Starts 12–14 years Ends beyond transfer at 18 years	Paediatric and adult cardiologist, nurses, psychologists, social workers, occupational therapists, physical therapists, speech and language therapists
Onofri, 2020 [27]	Long-term ventilation	Starts at early adolescence	Not specified

Downloaded from https://publications.ersnet.org on March 28, 2026 by guest. Please see licensing information on first page for reuse rights.

Complete remission is defined as an absence of clinical signs and symptoms, normal lung function and exercise tests and resolution of imaging abnormalities. However, it should be noted that this does not capture the dimension of risk of relapse.

Partial remission can consist of improvement of one or more of the aforementioned dimensions.

Remission has been reported spontaneously or during treatment and depends on the underlying disease. Diseases with a chance of spontaneous remission are nonfibrotic hypersensitivity pneumonitis after removal of the causative agent, NEHI, sarcoidosis and organising pneumonia [45–49]. For other chILD entities, partial remission can be achieved depending on underlying cause and treatment response. We summarise studies on chILD entities with long-term follow-up and (partial) remission.

These data confirm the necessity of continuous patient follow-up from childhood to adulthood. The risk of relapses later in life remains to be elucidated, especially in patients with apparently stable disease and prescribed no long-term treatment at the time of the transition; we do not have the tools for assessment of risk. Up to 50% of children with sarcoidosis still required treatment at the end of paediatric follow-up, the rest were stable off treatment. However, 19% of those stable patients relapsed later in adulthood. The authors concluded that childhood-onset sarcoidosis is characterised by a multivisceral involvement and a protracted course requiring follow-up into adulthood. Juvenile cases are characterised by severe organ manifestation and the severity cannot be predicted at onset [50].

Sisman *et al*. [51] followed 22 children with hypersensitivity pneumonitis for median of 3.3 years (age at end of follow-up ranging from 11.3 to 26.9 years) and most were stable off treatment. Six (27.3%) relapsed. Spirometry was normal in 90.9% of children at follow-up, while lung clearance index was increased in 47.4% cases, indicating persistent small airway disease. Abnormal total lung capacity, D_{LCO} and peak oxygen uptake were noted in 13.6%, 40.9% and 11.1%, respectively. Children with post-infectious bronchiolitis obliterans have been shown to suffer from severe airway obstruction, in some cases with need of supplemental oxygen. During a 12-year follow-up period, pulmonary function remained severely impaired, showing an obstructive pattern with air trapping. Although most of the patients stop needing supplemental oxygen around the age of 10 years, airway obstruction persisted with only slow if any improvement. Patients required frequent readmission due to recurrent respiratory infections; 19.6% developed chest deformity; and 15.2% underwent lobectomy because of treatment-resistant bronchiectasis, a higher figure than in most centres [52].

Diffuse alveolar haemorrhage (DAH) represents a heterogeneous group of conditions with variable onset and prognosis. In a large multicentric study of retrospective character Ring *et al*. [53] studied 117 children with DAH with a median (interquartile range (IQR)) age at presentation of 5 (2–12.9) years. There was a median (IQR) diagnostic delay of 2 (0–12) months. Outcome data were available for 90%, of whom 13% died (15 children); 86% were still alive: 19% were considered healthy, 16% were chronic clinically stable, 31% were active clinically stable and 12% had ongoing medical treatment and were unstable. While this study does not provide direct data on long-term follow-up into adulthood, the analysis of outcomes suggests that many such patients would be surviving into adulthood with sequelae of the disease and risk of a relapse.

ILD may complicate various rheumatological diseases and significantly impacts morbidity. Juvenile forms of rheumatological diseases are rare and poorly investigated with limited data on long-term follow-up. In a longitudinal follow-up of a cohort of 60 children with juvenile-onset systemic sclerosis from the European League Against Rheumatism (EULAR) Scleroderma Trials and Research (EUSTAR) database, the median age at disease onset was 12.4 years. Pulmonary fibrosis was present in 23.3% of patients and pulmonary hypertension in 13.3%. Median disease duration at last follow-up was 17.6 years and 26.6% of patients had active disease at that time. There were no data on the relapse rate in previously inactive disease [54].

ILD in surfactant protein C (SP-C) disease is very heterogeneous. Disease may manifest both in children and adults [55]. Follow-up of 17 children with SP-C disorder for a median 3 (range 0.9–19) years showed disease progression in three (17.6%) cases despite treatment, stability in seven (41.1%) patients and improvement in seven (41.1%) patients, one apparently becoming completely healthy. In a study of five children with open lung biopsy diagnosed ILD were followed long-term (mean 27.2 years) and all subsequently had confirmed *SFTPC* gene mutations. Three of the patients felt well after follow-up and had normal lung function; two females conducted normal activities of daily living, had healthy children but had evidence of restrictive lung disease. A positive effect of hydroxychloroquine seemed likely [56]. There are other papers on the evolution of health status in children with *SP-C* mutations [57–60]. It does appear that after years of relative stability there may be relapse and progression to refractory pulmonary fibrosis.

Downloaded from https://publications.ersnet.org on March 28, 2026 by guest. Please see licensing information on first page for reuse rights.

In adults, *SP-C* mutations may present at any age often with usual interstitial pneumonia and progressive pulmonary fibrosis (often no pulmonary disease is detected in childhood). Very recently a longitudinal study of *ABCA3*-related ILD in patients surviving beyond infancy demonstrated that this is a severe condition even in survivors beyond the first year of life and is characterised by radiological progression and functional decline [60].

Neuroendocrine cell hyperplasia of infancy and pulmonary interstitial glycogenosis are specific conditions of undefined aetiology usually presenting in infancy. Current data suggest favourable prognosis with symptom regression during early life [45, 61–64]. While 50% of patients with NEHI had no symptoms by age 3 years in one study [65], acute exacerbations have been documented in NEHI [63], and children with NEHI may also suffer from growth and developmental delay or asthma-like symptoms at school age which are not steroid-responsive [45, 65]. As these entities were not described until after 2000, we still lack data from adults regarding their long-term prognosis and risk of relapse.

In some other disease entities, a chronic progressive course has been seen with no documented remission. Mutations in *MARS* (methionyl-tRNA synthetase) manifest in infancy with pulmonary alveolar proteinosis and progress to fibrotic lung disease with chronic respiratory failure with a median survival of ~1 year [66, 67]. Hermansky–Pudlak syndrome type 2 manifests with fibrosing lung disease early in childhood (onset of respiratory symptoms on average at 5.5 years (range 0.8–12 years). At last follow-up at age 12 years (4–19 years) all patients had clinical and computed tomography abnormalities [68].

Question 5: Does the current definition of chILD versus *adult ILD influence transition of care?*

Literature search revealed that no study has been performed so far; therefore, only expert opinion dictates that lack of harmonisation between the current adult ILD and chILD classifications on transition of care affects transition negatively. A common understanding of terminology and disease states is a prerequisite for successful transfer. Based on the most recent classification, ILDs in adults can be broadly subgrouped into idiopathic, autoimmune-related, caused by external agents (*e.g.* exposure to organic and inorganic inhaled antigens or treatments, such as drugs or radiation), ILDs with cysts or airspace filling, sarcoidosis and orphan disorders [44]. The classification of chILD includes almost all the subgroups of the adult ILD classification in addition to specific entities that are absent from adult classifications including developmental disorders, such as alveolar capillary dysplasia, genetic surfactant disorders (of both synthesis and catabolism) and neuroendocrine cell hyperplasia of infancy. Furthermore, it embraces entities such as diffuse parenchymal lung disease (DPLD) related to lung vascular structural disease (*e.g.* pulmonary hypertension) and DPLD in the immunocompromised host or transplanted (bronchiolitis obliterans related to rejection) [69, 70]. There is no accepted age-overarching classification covering both paediatric and adult patients that could be routinely used in everyday care for the entire spectrum of chILD. A recent perspective, reviewing the history of ILD classifications, may be a good starting point for an international consensus to harmonise ILD disease classification [71].

In summary, important barriers based on differences between disease categorisations of children and adults could be addressed, to facilitate successful transition of chILD into adult ILD care.

Discussion

This ERS task force focused on mapping the availability of healthcare transition programmes specific for children with ILD reaching adulthood. While such programmes are available for some other chronic diseases and can help in the preparation of transition programmes for patients with chILD, no published specific transition programmes were found during our systematic literature search. Also, the survey performed during the COST Action found very few centres that had an established system, but even these centres varied in their routines and practices [25]. There was agreement within the task force on the suggestions that may help develop transition of care for children with chILD. The task force members discussed the possible variability of treatments between individual centres and agreed that there is a need of standardising the diagnostic and treatment protocols based on the available recommendations in the literature. As with any transition, it is a good plan for adult physicians to gain the confidence of the young person before making any treatment changes, unless this is absolutely essential. In the individual practice of the task force members there has been a major focus on children with neonatal or early-onset chILD. The task force members believe that there may be a substantial difference between the chILD with early childhood onset and the disease that occurs later. Most chILDs are diagnosed before the age of 2 years [69]. Many of these early-onset diseases either result in infant death or, for some diseases such as NEHI, may have a favourable prognosis with a high rate of remission. For these patients, this may interfere with the transition process as these patients (and their caregivers) may not feel the need for continuing medical follow-up and continuing care. However, there may be a high risk of relapse or late and progressive

Downloaded from https://publications.ersnet.org on March 28, 2026 by guest. Please see licensing information on first page for reuse rights.

respiratory complications in later life that could be detected early with continuing supervision. Adolescent-onset diseases may benefit from an easier transition process, as they are more likely to be symptomatic and still under treatment at the time of the transition process. Although these children may achieve long-term remission, the task force members agreed that the follow-up of these patients into adulthood help to better understand the long-term prognosis of child. The most important tool to learn more about the natural history of rare diseases including transition to adulthood and outcome, is including children and young people into national or international registries [1].

The European chILD registry (chILD-EU) has included >1000 patients with chILD [69]. It gives us an opportunity to generate data on transition. Approximately 250 chILD-EU patients are now aged >18 years. Approximately 30% were formally transferred to adult services; in ∼50% there was still some information coming from paediatric centres; in ∼10% there was no response; and 10% were lost to follow-up (Matthias Griese, Department of Pediatric Pneumology, Dr von Hauner Children's Hospital, Ludwig-Maximilians-University, German Center for Lung Research, Munich, Germany; personal communication). This clearly shows the need for enthusiasm of the treating physicians to continue reporting data to the registry [68]. Following chILD cases in the registry is a unique opportunity to obtain comprehensive knowledge of the disease across the developmental life-course, as previously realised for cystic fibrosis and many other conditions.

There are several barriers to effective transitional care. The most important ones identified by patients and their families are resistance of children and parents to transitioning (due to anxiety and fear of a new healthcare system and difficulty in leaving their paediatric clinicians); inadequate preparation and support for the transition process; and lack of coordination. The most common obstacles reported by paediatric and adult clinicians are lack of communication, coordination and sharing protocols between the paediatric and adult systems; lack of knowledge or training in paediatric-onset chronic diseases, and in adolescent development and behaviour. Other gaps reported are due to limited staff and financial barriers. Furthermore, although in some conditions, such as cystic fibrosis, there are excellent adult centres to which the patient can transition, in chILD there may be no adult physicians with expertise or interest.

The task force members agreed that it is important to overcome these barriers. Based on their clinical experience and current literature, the task force members felt that it would be useful to plan the implementation of a transition model and to engage both patients and parents. Important support may come from parents' groups and supporting organisations [24, 71]. Some reported aspects that can improve transition and outcomes are more time to address transition issues in clinic, support with additional staff, trained adult clinicians in paediatric-onset diseases and trained professionals in youth development [7, 11, 12, 26, 37]. Regarding the educational process of transition, different types of interventions have been published: transition preparation training delivered in a workshop, a web- and short message service (text)-based educational intervention, a nurse-led one-on-one teaching session or a structured comprehensive transition course. When a transition-oriented patient education programme has been conducted as a workshop or course, there is a positive effect on the competence of adolescents during the transition phase [14]. The task force members shared the opinion that despite a low number of patients, an established transition pathway from chILD centres to specialised centres for adult pulmonary rare diseases and use of the same transition centre-specific protocols shared across healthcare systems may have an added value. The opinion of the task force members was that patients with chILD would optimally be managed by doctors specialised in pulmonology/paediatric pulmonology with special expertise in the (ch)ILD. Rare pulmonary diseases require expertise with specialised pulmonary investigation methods and specific treatment options. The expertise of specialists in adolescent medicine who can address specific psychosocial problems related to transition or adolescence itself is also very helpful; for example, issues of substance abuse, social autonomy, sexuality or professional orientation, so there is a role for them in a multidisciplinary team.

Although there is a lack of evidence-based strategies for transition, studies have identified a variety of transition outcome variables. But to date, there is no agreement on which outcome variables should be measured, and there is a lack of more robust and consistent measurement of transition [39]. Future studies may incorporate all three components of transition (preparation, transfer and integration into adult care) in their design and evaluation of processes and outcomes and assess the impact of transition in terms of long-term outcomes of young adults.

In addition, there has been so far no political action at a national level with the authorities working with parents and children and young people, and healthcare professionals, in order to define strategies to implement policy.

Downloaded from https://publications.ersnet.org on March 28, 2026 by guest. Please see licensing information on first page for reuse rights.

TABLE 5 Identified research and clinical gaps

Lack of child transition protocols
Lack of communication between paediatric and adult centres.
Lack of coordination
Lack of knowledge and training of involved healthcare professionals
Lack of expertise in chILD in adult centres and in adult ILD in chILD centres
Deficit of targeted financial support
Need for research on long-term prognosis of early-onset chILDs

chILD: childhood interstitial lung disease; ILD: interstitial lung disease.

Table 5 summarises identified gaps that need to be filled, both in clinical practice and in research.

Conclusions

- chILDs are often different to ILDs in adults.
- Transition of care programmes for children with chILD are lacking.
- Knowledge about chILD in adult centres is insufficient.
- There is a lack of harmonisation between the current adult ILD and chILD classifications.
- Enhanced efforts are necessary to include chILD patients into registries for long-term follow-up of children from early-onset chILD into adulthood.
- There is no structured, internationally utilised system of long-term follow-up of children with chILD.
- There is a substantial deficit of financial support and personnel for the transition process.

This statement summarises the current knowledge regarding the long-term follow-up of children with chILD and the transition of care. It describes the need of standardised protocols and multidisciplinary approach with good communication between the paediatric and adult centres.

This document was endorsed by the ERS Executive Committee on 15 May 2024.

Acknowledgement: The authors sincerely thank Thomy Tonia (Institute of Social and Preventive Medicine, University of Bern, Bern, Switzerland), the methodologist of the European Respiratory Society, for her major help and support. The authors also thank Jeanette Boyd (European Lung Foundation, Sheffield, UK) for reviewing and commenting on the manuscript.

Author contributions: All authors contributed to this manuscript in terms of authorship requirements as published in the *European Respiratory Journal* manuscript preparation guidelines. All authors approved this final version of the manuscript.

Conflict of interest: P. Pohunek reports consulting fees, travel support and advisory board participation with AstraZeneca and GlaxoSmithKline, and lecture honoraria from GlaxoSmithKline, Chiesi and AstraZeneca; outside the submitted work. E. Manali reports lecture honoraria from Boehringer Ingelheim, Elpen, Demo, CSL Behring and Hoffman La Roche, and travel support from Boehringer Ingelheim, Hoffman La Roche, Elpen and CSL Behring, outside the submitted work. F. Chua reports lecture honoraria and travel support from Boehringer Ingelheim, and advisory board participation with Boehringer Ingelheim and the National Institute for Care Excellence (NICE), UK, outside the submitted work. R. Epaud reports consulting fees from AstraZeneca, lecture honoraria from GSK, AstraZeneca and Menarini, travel support from GSK, and AstraZeneca, and advisory board participation with AstraZeneca and Novartis, outside the submitted work. C. Gilbert reports a leadership role with ChILD Lung Foundation, outside the submitted work. M. Griese reports grants, lecture honoraria, advisory board participation and adjudication board participation with Boehringer Ingelheim, outside the submitted work. V. Koucký reports advisory board participation with Boehringer Ingelheim, outside the submitted work. N. Nathan reports grants from Million Dollar Bike Ride project (for Neuroendocrine Cell Hyperplasia of Infancy: Genetic basis of neuroendocrine cell hyperplasia of infancy), Chancellerie des Universités: Legs Poix (Molecular and phenotypic characterisation of interstitial lung disease, number 2022000594) and RespiFIL (development of an e-learning module for CT-scan in childhood interstitial lung diseases and development of an online platform for the collection of quality of life and transition questionnaires in rare lung disease), lecture honoraria from La lettre du Pneumologue, and travel support from ERS, outside the submitted work. S. Papiris reports lecture honoraria from Boehringer Ingelheim, Elpen, Demo, Pfizer and Hoffman La Roche, and travel support from Boehringer Ingelheim and Elpen. S. Terheggen-Lagro reports advisory board participation with Roche, outside the submitted work. All other authors have no potential conflicts of interest to disclose.

Downloaded from https://publications.ersnet.org on March 28, 2026 by guest. Please see licensing information on first page for reuse rights.

Support statement: This work was supported by the European Respiratory Society (ERS Task Force TF-2021-14). Funding information for this article has been deposited with the Crossref Funder Registry.

References

1 Griese M. Chronic interstitial lung disease in children. *Eur Respir Rev* 2018; 27: 170100.
2 Bush A, Gilbert C, Gregory J, *et al.* Interstitial lung disease in infancy. *Early Hum Dev* 2020; 150: 105186.
3 Spagnolo P, Bush A. Interstitial lung disease in children younger than 2 years. *Pediatrics* 2016; 137: e20152725.
4 Torrent-Vernetta A, Gaboli M, Castillo-Corullón S, *et al.* Incidence and prevalence of children's diffuse lung disease in Spain. *Arch Bronconeumol* 2022; 58: 22–29.
5 Breuer O, Schultz A. Side effects of medications used to treat childhood interstitial lung disease. *Paediatr Respir Rev* 2018; 28: 68–79.
6 Gappa M, Bush A, Ersu RH, *et al.* Children and the European Respiratory Society: from silos to synergies. *Eur Respir J* 2023; 62: 2301692.
7 Mittaine M, Roditis L, Dupuis M. Transition des patients atteints d'une maladie respiratoire chronique depuis la pédiatrie vers les services pour adultes: l'exemple de la mucoviscidose. [Transition from pediatric to adult care in chronic respiratory diseases: the cystic fibrosis model]. *Med Sci* 2023; 39: 58–63.
8 Buschur EO, Glick B, Kamboj MK. Transition of care for patients with type 1 diabetes mellitus from pediatric to adult health care systems. *Transl Pediatr* 2017; 6: 373–382.
9 Bassareo PP, Chessa M, Di Salvo G, *et al.* Strategies to aid successful transition of adolescents with congenital heart disease: a systematic review. *Children* 2023; 10: 423.
10 Ross J, Bowden MR, Yu C, *et al.* Transition of young adults with metabolic bone diseases to adult care. *Front Endocrinol* 2023; 14: 1137976.
11 Vandergraesen T, Dedeken L, Efira A, *et al.* Successful transition from paediatric to adult care in sickle cell disease. *Rev Med Brux* 2019; 40: 495–501.
12 Philpott JR, Kurowski JA. Challenges in transitional care in inflammatory bowel disease: a review of the current literature in transition readiness and outcomes. *Inflamm Bowel Dis* 2019; 25: 45–55.
13 Moons P, Bratt E-L, De Backer J, *et al.* Transition to adulthood and transfer to adult care of adolescents with congenital heart disease: a global consensus statement. *Eur Heart J* 2021; 42: 4213–4223.
14 Toulany A, Gorter JW, Harrison M. A call for action: recommendations to improve transition to adult care for youth with complex health care needs. *Paediatr Child Health* 2022; 27: 297–309.
15 Huang JS, Terrones L, Tompane T, *et al.* Preparing adolescents with chronic disease for transition to adult care: a technology program. *Pediatrics* 2014; 133: e1639–e1646.
16 Schmidt S, Herrmann-Garitz C, Bomba F, *et al.* A multicenter prospective quasi-experimental study on the impact of a transition-oriented generic patient education program on health service participation and quality of life in adolescents and young adults. *Patient Educ Couns* 2016; 99: 421–428.
17 Bourgeois G, Magne F, Nove Josserand R, *et al.* A formalized transition program for cystic fibrosis: a 10-year retrospective analysis of 97 patients in Lyon. *Pediatr Pulmonol* 2021; 56: 2000–2006.
18 Welsner M, Sutharsan S, Taube C, *et al.* Changes in clinical markers during a short-term transfer program of adult cystic fibrosis patients from pediatric to adult care. *Open Respir Med J* 2019; 13: 11–18.
19 Peeters MAC, Sattoe JNT, van Staa AL, *et al.* Controlled evaluation of a transition clinic for Dutch young people with cystic fibrosis. *Pediatr Pulmonol* 2019; 54: 1811–1820.
20 Skov M, Teilmann G, Damgaard IN, *et al.* Initiating transitional care for adolescents with cystic fibrosis at the age of 12 is both feasible and promising. *Acta Paediatr* 2018; 107: 1977–1982.
21 Chaudhry SR, Keaton M, Nasr SZ. Evaluation of a cystic fibrosis transition program from pediatric to adult care. *Pediatr Pulmonol* 2013; 48: 658–665.
22 Tuchman L, Schwartz M. Health outcomes associated with transition from pediatric to adult cystic fibrosis care. *Pediatrics* 2013; 132: 847–853.
23 Middour-Oxler B, Bergman S, Blair S, *et al.* Formal *vs*. informal transition in adolescents with cystic fibrosis: a retrospective comparison of outcomes. *J Pediatr Nurs* 2022; 62: 177–183.
24 Craig SL, Towns S, Bibby H. Moving on from paediatric to adult health care: an initial evaluation of a transition program for young people with cystic fibrosis. *Int J Adolesc Med Health* 2007; 19: 333–343.
25 Koucký V, Pohunek P, Vašáková M, *et al.* Transition of patients with interstitial lung disease from paediatric to adult care. *ERJ Open Res* 2021; 7: 00964-2020.
26 Spielberg DR, Weinman J, DeBoer EM. Advancements in imaging in ChILD. *Pediatr Pulmonol* 2023; in press [https://doi.org/10.1002/ppul.26487].
27 Onofri A, Broomfield A, Tan HL. Transition to adult care in children on long-term ventilation. *Front Pediatr* 2020; 8: 548839.
28 Calvo I, Antón J, Bustabad S, *et al.* Consensus of the Spanish society of pediatric rheumatology for transition management from pediatric to adult care in rheumatic patients with childhood onset. *Rheumatol Int* 2015; 35: 1615–1624.

Downloaded from https://publications.ersnet.org on March 28, 2026 by guest. Please see licensing information on first page for reuse rights.

29 Canadian Association of Paediatric Health Centres. A Guideline for Transition from Paediatric to Adult Health Care for Youth with Special Health Care Needs: A National Approach. 2016. https://www.childhealthbc.ca/sites/default/files/caphc_transition_to_adult_health_care_guideline_may_2017.pdf

30 Foster HE, Minden K, Clemente D, *et al.* EULAR/PReS standards and recommendations for the transitional care of young people with juvenile-onset rheumatic diseases. *Ann Rheum Dis* 2017; 76: 639–646.

31 National Institute for Health and Care Excellence (NICE). Transition from Children's to Adults' Services for Young People Using Health or Social Care Services. NICE guideline [NG43]. 2016. www.nice.org.uk/guidance/ng43. Date last updated: 24 February 2016.

32 Cooley WC, Sagerman PJ, Barr MS, *et al.* Supporting the health care transition from adolescence to adulthood in the medical home. *Pediatrics* 2011; 128: 182–200.

33 White PH, Cooley WC. Supporting the health care transition from adolescence to adulthood in the medical home. *Pediatrics* 2018; 142: e20182587.

34 McManus M, White P, Barbour A, *et al.* Pediatric to adult transition: a quality improvement model for primary care. *J Adolesc Health* 2015; 56: 73–78.

35 Goldenring BJM, Rosen DS. Getting into adolescent heads: an essential update. *Contemp Pediatr* 2004; 21: 64–90.

36 Nagra A, McGinnity PM, Davis N, *et al.* Implementing transition: ready steady go. *Arch Dis Child Educ Pract Ed* 2015; 100: 313–320.

37 Cook J, Chitty LS, De Coppi P, *et al.* The natural history of prenatally diagnosed congenital cystic lung lesions: long-term follow-up of 119 cases. *Arch Dis Child* 2017; 102: 798–803.

38 Bolger A, Vargus-Adams J, McMahon M. Transition of care in adolescents with cerebral palsy: a survey of current practices. *PM R* 2017; 9: 258–264.

39 Gabriel P, McManus M, Rogers K, *et al.* Outcome evidence for structured pediatric to adult health care transition interventions: a systematic review. *J Pediatr* 2017; 188: 263–269.

40 Campbell F, Biggs K, Aldiss SK, *et al.* Transition of care for adolescents from paediatric services to adult health services. *Cochrane Database Syst Rev* 2016; 2016: CD009794.

41 Gilbert C, Bennett KM, Brown C, *et al.* Experiences of UK-based adult transition services for interstitial lung disease in childhood: "There's a lot less cushioning". *Pediatr Pulmonol* 2023; 58: 1993–1999.

42 White P, Schmidt A, Shorr J, *et al.* Six Core Elements of Health Care Transition™ 3.0: transitioning to an adult approach to health care without changing clinicians. Washington, DC: Got Transition, The National Alliance to Advance Adolescent Health. 2020. www.gottransition.org/6ce/?Staying-Full-Package.

43 Singh J, Towns S, Jayasuriya G, *et al.* Transition to adult care in cystic fibrosis: the challenges and the structure. *Paediatr Respir Rev* 2022; 41: 23–29.

44 Wijsenbeek M, Suzuki A, Maher TM. Interstitial lung diseases. *Lancet* 2022; 400: 769–786.

45 Lukkarinen H, Pelkonen A, Lohi J, *et al.* Neuroendocrine cell hyperplasia of infancy: a prospective follow-up of nine children. *Arch Dis Child* 2013; 98: 141–144.

46 Dervaux M, Thumerelle C, Fabre C, *et al.* Long-term evolution of neuroendocrine cell hyperplasia of infancy: the FRENCHI findings. *Eur J Pediatr* 2023; 182: 949–956.

47 Buchvald F, Petersen BL, Damgaard K, *et al.* Frequency, treatment, and functional outcome in children with hypersensitivity pneumonitis. *Pediatr Pulmonol* 2011; 46: 1098–1107.

48 Lopes MC, Amadeu TP, Ribeiro-Alves M, *et al.* Defining prognosis in sarcoidosis. *Medicine* 2020; 99: e23100.

49 Abi-Khalil S, Fadel N, Menassa-Moussa L, *et al.* Spontaneous resolution of cryptogenic organizing pneumonia in pediatrics: a case report. *Arch Pediatr* 2016; 23: 519–522.

50 Chauveau S, Jeny F, Montagne ME, *et al.* Child–adult transition in sarcoidosis: a series of 52 patients. *J Clin Med* 2020; 9: 2097.

51 Sisman Y, Buchvald F, Blyme AK, *et al.* Pulmonary function and fitness years after treatment of biopsy verified hypersensitivity pneumonitis in childhood. *Am J Respir Crit Care Med* 2015; 191: A5458.

52 Colom AJ, Maffey A, Garcia Bournissen F, *et al.* Pulmonary function of a paediatric cohort of patients with postinfectious bronchiolitis obliterans. A long term follow-up. *Thorax* 2015; 70: 169–174.

53 Ring AM, Schwerk N, Kiper N, *et al.* Diffuse alveolar haemorrhage in children: an international multicentre study. *ERJ Open Res* 2023; 9: 00733-2022.

54 Foeldvari I, Tyndall A, Zulian F, *et al.* Juvenile and young adult-onset systemic sclerosis share the same organ involvement in adulthood: data from the EUSTAR database. *Rheumatology* 2012; 51: 1832–1837.

55 Kröner C, Reu S, Teusch V, *et al.* Genotype alone does not predict the clinical course of *SFTPC* deficiency in paediatric patients. *Eur Respir J* 2015; 46: 197–206.

56 Avital A, Hevroni A, Godfrey S, *et al.* Natural history of five children with surfactant protein C mutations and interstitial lung disease. *Pediatr Pulmonol* 2014; 49: 1097–1105.

57 Thouvenin G, Taam RA, Flamein F, *et al.* Characteristics of disorders associated with genetic mutations of surfactant protein C. *Arch Dis Child* 2010; 95: 449–454.

58 Nogee LM, Dunbar AE, Wert SE, *et al.* A mutation in the surfactant protein C gene associated with familial interstitial lung disease. *N Engl J Med* 2001; 344: 573–579.

Downloaded from https://publications.ersnet.org on March 28, 2026 by guest. Please see licensing information on first page for reuse rights.

59 Taam RA, Jaubert F, Emond S, *et al.* Familial interstitial disease with I73T mutation: a mid- and long-term study. *Pediatr Pulmonol* 2009; 44: 167–175.

60 Kazzi B, Lederer D, Arteaga-Solis E, *et al.* Recurrent diffuse lung disease due to surfactant protein C deficiency. *Respir Med Case Rep* 2018; 25: 91–95.

61 Gomes VCC, Silva MCC, Maia Filho JH, *et al.* Diagnostic criteria and follow-up in neuroendocrine cell hyperplasia of infancy: a case series. *J Bras Pneumol* 2013; 39: 569–578.

62 Seidl E, Carlens J, Reu S, *et al.* Pulmonary interstitial glycogenosis – a systematic analysis of new cases. *Respir Med* 2018; 140: 11–20.

63 Houin PR, Deterding RR, Young LR. Exacerbations in neuroendocrine cell hyperplasia of infancy are characterized by increased air trapping. *Pediatr Pulmonol* 2016; 51: E9–E12.

64 Canakis AM, Cutz E, Manson D, *et al.* Pulmonary interstitial glycogenosis: a new variant of neonatal interstitial lung disease. *Am J Respir Crit Care Med* 2002; 165: 1557–1565.

65 Nevel RJ, Garnett ET, Schaudies DA, *et al.* Growth trajectories and oxygen use in neuroendocrine cell hyperplasia of infancy. *Pediatr Pulmonol* 2018; 53: 656–663.

66 Enaud L, Hadchouel A, Coulomb A, *et al.* Pulmonary alveolar proteinosis in children on La Réunion island: a new inherited disorder? *Orphanet J Rare Dis* 2014; 9: 85.

67 Hadchouel A, Wieland T, Griese M, *et al.* Biallelic mutations of methionyl-tRNA synthetase cause a specific type of pulmonary alveolar proteinosis prevalent on Réunion island. *Am J Hum Genet* 2015; 96: 826–831.

68 Hengst M, Naehrlich L, Mahavadi P, *et al.* Hermansky-Pudlak syndrome type 2 manifests with fibrosing lung disease early in childhood. *Orphanet J Rare Dis* 2018; 13: 42.

69 Griese M, Seidl E, Hengst M, *et al.* International management platform for children's interstitial lung disease (chILD-EU). *Thorax* 2018; 73: 231–239.

70 Nathan N, Griese M, Michel K, *et al.* Diagnostic workup of childhood interstitial lung disease. *Eur Respir Rev* 2023; 32: 220188.

71 Griese M. Etiologic classification of diffuse parenchymal (interstitial) lung diseases. *J Clin Med* 2022; 11: 1747.

Downloaded from https://publications.ersnet.org on March 28, 2026 by guest. Please see licensing information on first page for reuse rights.